CURRICULUM VITAE

Hao-Jui Kuan

Date of birth : 13 Jul 1995
Nationality : Taiwan
Email/web : hjkuan@illinois.edu / **hao-jui.github.io**

Professional experience

2025 – Present Research scholar at Illinois Center for Advanced Studies of the Universe & University of Illinois Urbana-Champaign, United States. *PI: Nicolas Yunes, Jacquelyn Noronha-Hostler.*

2023 – 2025 Postdoc in the Computational Relativistic Astrophysics division at the Max Planck Institute for Gravitational Physics (Albert Einstein Institute), Germany. *PI: Masaru Shibata.*

Education

2019 – 2022 University of Tübingen, Tübingen, Germany. *Advisor: Kostas D. Kokkotas*
PhD in Theoretical Astrophysics. Summa Cum Laude (excellent; $\lesssim 1$ in 50 graduates).

2017 – 2022 National Tsing Hua University, Taiwan. *Advisor: Chao-Qiang Geng*
PhD in Physics (joint degree).

2013 – 2017 National Tsing Hua University, Taiwan.
BSc Double major in Physics and Mathematics.

Honors and Awards

2023 Honorable Mention in the GWIC- **Braccini Thesis Prize**

2023 Dr. Friedrich Förster Prize, one awardee in Faculty of Science in University of Tübingen, €1,000.

2023 Student Outstanding Paper Award from NCTS (Taiwan), converted as $\sim$ 620 USD.

Fellowships and Grants

2020 – 2021 **[PI]** "Neutron stars as gravitational wave sources", Sandwich-Scholarship Programme, DAAD and MOST (Taiwan), €13,500.

2017 – 2020 Presidential Scholarship of National Tsing Hua University (Taiwan), converted as ∼18,700 USD.

Reviewing Activities

Journals Physical Review Letter, Physical Review D, International Journal of Modern Physics D, European Physical Journal C, Monthly Notices of the Royal Astronomical Society

List of Presentations

Plenary Talks and Colloquia

2025

Feb 11 "**Uncovering Extreme Matter and Gravity through Gravitational Waves from Neutron Star Mergers**", Academia Sinica, Taipei, Taiwan.

2022

Sep 10 "**Premerger Neutron Star Physics**", Eleventh Aegean summer school, Syros, Greece.

Invited Talks

2026

Nov 5 "**Donutization: When Massive Scalar Fields Hide Inside Neutron Stars**", Gravity seminar at Perimeter Institute, Toronto, Canada.

Sep 18 "**Neutron-star oscillations as a spectroscopic probe of dense nuclear physics with gravitational waves**", Taiwan-Korea Nuclear Physics, Academia Sinica, Taipei, Taiwan.

2025

Oct 8 "**Tidal Resonance: Nonlinearities revealed by numerical relativity**", Astrophysics, Relativity, and Cosmology Seminar at UIUC.

Jun 17 "**Tidal Resonance in Binary Neutron Star Mergers – Numerical Relativity Insights**", Conference website, Varna, Bulgaria.

Feb 27 "**Numerical Relativity Insights into Tidal Resonance Phenomena**", University of Tübingen, Germany.

2024

Nov 25 "**Probing Subatomic Physics and Gravity through Gravitational Waves from Neutron Star Mergers**", Academia Sinica, Taipei, Taiwan.

Aug 1 "**On the finite-size imprints on waveforms of binary neutron star mergers**", UIUC.

Jul 11 "**Imprints of matter and scalar effects on waveforms from binary neutron star mergers**", online-seminar-in-mathematical-numerical-relativity (website).

Apr 24 "**Imprints of matter and scalar effects on waveforms from binary neutron star mergers**", Institute for Gravitational and Subatomic Physics (GRASP), Utrecht University.

Apr 8 "**(Dynamical) Tidal Effects in the Binary Neutron Star mergers**", AstroParticle and Cosmology laboratory (APC), Paris, France.

2023

Mar 16 "**Dynamical Scalarization during Neutron Star Mergers in scalar-Gauss-Bonnet Theory**", University of Tübingen, Germany.

Feb 3 "**Developing waveform involving dynamical tides**", Academia Sinica, Taipei, Taiwan.

Feb 2 "**Gravitational Phase Transition in Massive Scalar-tensor Theory**", National Center for Theoretical Sciences, Hsinchu, Taiwan.

2021

Jul 27 "g**-mode resonances as triggers for precursors of SGRBs**", Grav@zon group, Federal University of Pará, Pará, Brazil.

Jun 22 "**Dynamical formation of scalarized black holes and neutron stars through stellar core collapse**", cosmo/GW journal club, Johns Hopkins University, Maryland, USA.

Contributed Talks

2026

Oct 9 "**Shell-Localized Scalar Fields in Neutron Stars: How They Affect the Interior**", Midwest Relativity Meeting, Oakland University, Michigan, USA.

Feb 5 "**Tidal Resonance: Nonlinearities revealed by numerical relativity**", Multi-Messenger Astrophysics in the Dynamic Universe, YITP, Kyoto, Japan.

2025

Jul 24 "**Tidal Resonance in Binary Neutron Star Mergers**", New Frontiers 2025 in Numerical Relativity, Mallorca, Spain.

2024

Oct 9 "**On the finite-size imprints on waveforms of binary neutron star mergers**", Observatoire de Paris - site de Meudon, France.

Feb 28 "**Binary neutron star mergers in massive scalar-tensor theory**", Gravity and Cosmology 2024, Yukawa Institute for Theoretical Physics, Kyoto, Japan.

2023

Oct 26 "**Binary neutron star mergers in massive scalar-tensor theory: an adiabatic look**", poster, GravityShapePisa 2023, Pisa, Italy.

Apr 5 "**Dynamical Scalarization during BNS mergers in scalar-Gauss-Bonnet**", CoCoNut meeting, Potsdam, Germany.

Mar 22 "**Packed Message delivered by Tides in Binary Neutron Star Mergers**", SMuK2023, Technical University Dresden, Germany.

2022

Sep 8 "**Gravitational Phase Transition**", Eleventh Aegean summer school, Syros, Greece.

May 16–19 "**Tidal effects in the pre-merger stage of coalescing binary neutron stars**", e-poster, PHAROS: The multi-messenger physics and astrophysics of neutron stars, Roma, Italy.

Mar 3 "**Resonance shattering as triggers for precursors of SGRBs**",
DPG Meeting of the Matter and Cosmos section (SMuK), Heidelberg, Germany.

2021

Aug 30 – Sep 3 "**Tidal g-mode resonances in coalescing binaries of neutron stars as triggers for precursor flares of short gamma-ray bursts**",
DPG Meeting of the Matter and Cosmos section (SMuK), Bad Honnef, Germany.

Feb 24–28 "**Inverse-Chirp Imprint of GW in Scalar Tensor Theory**",
56th Karpacz Winter School in Theoretical Physics, Karpacz, Poland.

Advising and Mentoring

Current Graduate Students

2025 – Present Carlos Conde, UIUC, co-advised with Prof. Yunes.
2025 – Present Daniel Caballero, UIUC, co-advised with Prof. Yunes.
2025 – Present Zoey Dong, UIUC, co-advised with Prof. Yunes.

Teaching Activities

Teaching assistant

2016 – 2019 *Tsing Hua:* General Relativity (2016, 2018 Fall), Mathematical Physics (2017–18), Thermal Physics (2016–17), Calculus (2018–19)
2021 – 2022 *Tübingen:* Physical Practical (2021 Summer & Winter), Grundkurs Electromagnetism (2022 Summer)

Outreach

Mar 2025 Black hole lectures, *Jürgen Ehlers Spring School*, Potsdam.

Computer Skills and Languages

Programming Python, Fortran, C++
Scientific software Matlab, Mathematica, Gnuplot
Codes **GRASS** (General Relativistic Axisymmetric Spacetime Solver)
Celephais (Kadath-based numerical-relativity code)